A Political Economy Model of the Chip War: Supply Chain Weaponization and Its Long-term Consequences

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Abstract

When can states successfully attack others with trade restrictions? This paper explores strategic trade restrictions on products that require high technology and complex supply chains. I show that trade restrictions incentivize less productive firms to enter the market who have not been able to and may ironically benefit the targeted firm, unless its original trade partner strikes a collusive negotiation with the potential competitors or the government persuades other countries to join the restrictions. The resulting shift in the trade network is not necessarily easy to reverse. Therefore, states should take into account firms' strategic responses to ensure that the restrictions exert pressure on the target as intended. My results demonstrate how important it is to consider the economic outcome for the success of trade restrictions and how they have become an increasingly limited policy tool as production gets more fragmented.

Keywords: trade restriction, supply chains, technology, semiconductors, weaponized interdependence, formal model

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1 Introduction

Semiconductors are at the center of the crossfire between the world's two biggest economies (Bown, 2020). As China has grown into a behemoth of the industry that has enormous economic, strategic and military values, the U.S. is trying to cut off the economic ties with extraordinarily strong measures in hope of defending its technological supremacy. The ongoing chip war warrants attentions not only because of its practical implications. A group of scholars have recently studied the patterns of how states weaponize interdependence as a coercive diplomacy in wide areas ranging from security to trade to finance to energy (Drezner, Farrell and Newman, 2021; Farrell and Newman, 2019). The modern semiconductor industry makes a strong test case of their theories since no other industries have been involved in the trade network as complex and the American firms are not welcoming trade restrictions, unlike the last chip war between the U.S. and Japan (Friedberg, 1991). If states insist on restrictions against business interests, do firms no longer hold any relevance to trade politics?

The current literature mostly views the strategic competition between the U.S. and China from the economic statecraft perspective, a tradition that emphasizes states' autonomy (Aggarwal and Reddie, 2021; Weiss, 2021). It argues the economic dimension is less important to the success of sanctions than their political configuration such as their international regime and the political signal they convey (Baldwin, 2020; Drezner, 1999). However, restrictive measures are pointless if they ultimately benefit the target. Japan banned exports of crucial chipmaking materials to Korea in 2019 that no others could produce at that time, only to help Korean firms secure alternative sources and undermine its own monopoly. Firms can sidestep restrictive efforts of states with strategic investment in the world of fragmented production (Beazer and Blake, 2018; Betz and Pond, 2019) and trade restrictions are no exception. I claim that the firm-level approach of political economy can contribute to a more balanced view of trade restrictions. This paper specifically examines firms' strategic responses to restrictions and their consequences on supply chains.[1]

[1]This paper relates to several strands of the most recent political economy literature. Studies on the U.S.-China trade war (Fajgelbaum and Khandelwal, 2021; Kim and Margalit, 2021) and the public backlash against the liberal international order (Lake, Martin and Risse, 2021; Mansfield, Milner and Rudra, 2021) provide the starting point for our discussion, from which this paper points to one of the directions those phenomena are leading to. Theories on technology and politics (Milner and Solstad, 2021) and security

The key idea is that the trade network is more flexible than the network of states. Nodes can be newly introduced (market entry) or converge (collusion) with ease in the trade network. As a result, shocks as extreme as banning trade can cause a seismic shift of the status quo, violating the fundamental assumption of weaponized interdependence theories that networks are embedded in a contained system (Farrell and Newman, 2019). Firms face no incentives to automatically revert to the original relationships once adjustment costs are paid, so the resulting shift in the network, big or small, may not be easy to reverse even when the restrictions are lifted. It also does not help that the degree that firms are vulnerable to restriction often depends on private information that states cannot easily come by such as their detailed investment plans and the availability of alternative trade partners. The combination of the market force and the information asymmetry renders trade restriction an increasingly limited policy tool as economic interdependence deepens.

I present a political economy model inspired by the Korea-Japan trade row of 2019 to support the arguments. In the model, a government bans its firm from selling certain goods to a target firm in the model. The targeted firm in turn seeks alternative suppliers that offer imperfect substitutes for a lower price with both making some initial investments. The equilibrium analysis states that trade restrictions incentivize less productive firms to enter the market who have not been able to so far. If the market competition has increased as they entered during the restriction, the exporter earns less and its trade partners earn more than what each had earned before, possibly a blessing in disguise for the targeted firm. The competitors do not enter if the exporter strikes a collusive negotiation through partnership or mergers with them or the government persuades other countries to join the restrictions. The two have divergent preferences over the options of collusion and multilateral restrictions, but the government has few economic interests in deterring the exporter from continuing operation when it fails to mobilize other countries' participation.

and trade (Davis and Pratt, 2021) can be tested with the unfolding of the semiconductor war this paper suggests. Our study overlaps with: Cory, Lerner and Osgood (2021) and Malesky and Milner (2021) in that both study supply chains and politics, but they examine the political outcomes of economic change while we examine the economic consequences of political change; Cohen and Rogers (2021) and Wu (2021) in that both study supply chains and trade restrictions, but they examine their legal development while we examine their distributional outcome; Lim and Ferguson (Forthcoming) in that both study the outcome of trade restrictions, but their main explanatory variable is formality of the measures while ours is trade restriction in general.

The model has theoretical and practical implications. Theoretically, it delimits the extent to which states can weaponize economic interdependence. States have to take into account strategic responses of both domestic and foreign firms to ensure the restrictions exert pressure on another state as intended. They not only suffer long-term economic losses but lose the opportunity to weaponize the same economic ties for good when things go awry. They might be willing to tolerate losing some money for their political goals. However, here comes the paradox: for trade restrictions to be effective, the state have to sell the idea of restrictions to the very countries that are most likely to reap economic benefits by trading with the targeted country. Supply chains can be viewed as a prisoner's dilemma in that unilateral withdrawal benefits others at the cost of the state itself without effectively dismantling the supply chains even if state leaders wish to walk away from them. The model predicts that the ongoing chip war is an exception, not an indicator of paradigm shifts. These results can be extended from semiconductors to all products whose production requires technology and flexible supply chains including batteries and pharmaceutics, the two products specially noted by the White House (2021).

Practically, the results easily relate to the ongoing chip war. The U.S. is trying to multilateralize trade restrictions against China by doling out side payments and constructing supply chains centered around its allies. This model stresses the importance of securing alternative trade partners for both the U.S. and China. Once the U.S. secures alternative manufacturers free from Chinese influence, American firms face less incentives to restore trade with China; should it happen, Chinese firms have to endure smaller margins. Conversely, trade restrictions lose some relevance if China serves new customers that are free from American influence, most likely its own emerging tech firms. The collusive solution against China from the private sector is unlikely due to the rapid technological development and the fierce market competition, which makes the role of states all the more important. Therefore, failure to secure alternative supply chains before China does would inflict a significant long-term political cost on the U.S. government.

This paper is organized as follows: Section 2 presents the formal specification of the model and the equilibrium analysis. Section 3 concludes the paper with discussion.

2 Model

2.1 Settings

Usual trade flow data only include firms that are currently participating in trade. However, to fully capture the ramifications of trade restriction, we need to count in latent players that have a potential to enter the market once restriction is enacted but are yet to. These potential competitors often produce similar or related products whose technological contents are not sophisticated enough compared to the incumbent players. In general, more advanced technology better suits producers. Firms invest to outdo their competitors, not only exporters who produce them but also importers who have to adjust their production lines to accommodate the better inputs. Affected firms look for alternative partners upon restriction and trade with them if it suits their interest. They decide to continue the relationship or revert to the original partners when the restriction is lifted.

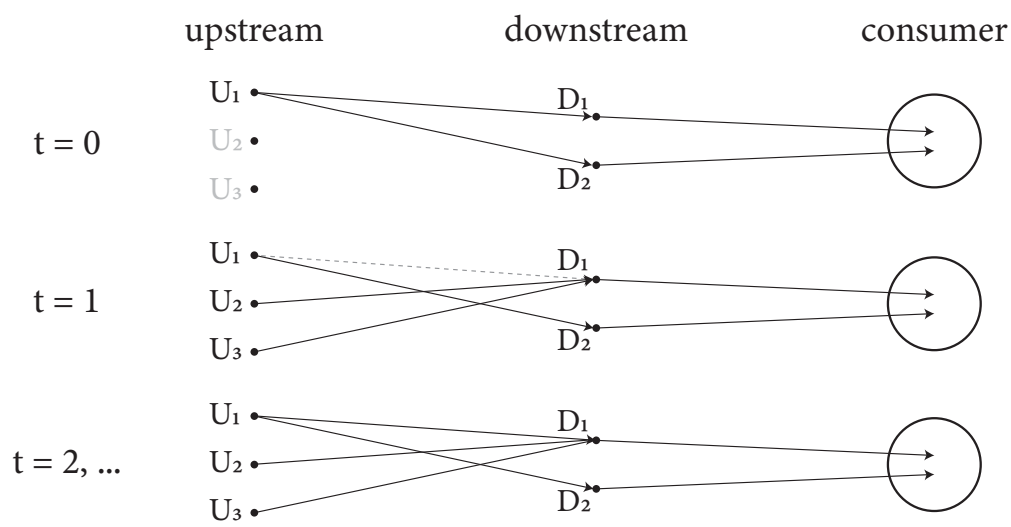

Figure 1: An example of a change in the trade network under export restriction: U_1 is a monopolistic exporter and D_1 and D_2 are its customers. When U_1's government imposes export restriction on D_1 at $t = 1$, the removal of the restriction at $t = 2$ does not ensure that D_1 trades back with U_1 exclusively.

Figure 1 illustrates one example of an export restriction model. We abstract away from other than the two industries that are directly involved in the restrictive measure and only focus on the partial equilibrium among the two types of firms and consumers. Upstream

producers U_is and downstream producers D_js each refer to firms that export and import the restricted goods. U_1 is the only supplier for downstream producers at $t = 0$, successfully fending off latent competitors U_2 and U_3. Suppose that U_1's government abruptly imposes trade restriction on D_1 at $t = 1$ as depicted in the severed tie between them. D_1 has an incentive to invite one of U_2 and U_3 into the market in order not to forfeit the entire profit during restriction. Although not depicted in the figure, U_1 might also seek alternative downstream producers who bid at lower prices than D_1 but would bring in some profits anyway. We assume the restriction lasts for one period and adjust the value of the time discount β accordingly to its duration.

It takes a massive effort to construct an overarching micro model that accounts for all the individual differences enumerated in Appendix A. I instead conduct a Nash equilibrium analysis for a model that specifically caters to the Korea-Japan trade row of 2019 whose insights are nevertheless generalizable. The model basically follows that in Figure 1 but with several more simplifying assumptions. We impose symmetry between the downstream producers (D_1 and D_2) and between the latent competitors (U_2 and U_3). Investment is made only at the beginning of $t = 1$ and takes effect immediately. Technology has an intrinsic value only and is unidimensional. Most importantly, the upstream producer U_1 faces zero production cost so it will not look for alternative downstream producers upon restriction.[2]

We first state each player's goal at $t = 0$ that characterizes the pre-restriction equilibrium. Suppose U_1 produces at sophistication level $s_1 = \bar{s}$ and latent competitors U_2 and U_3 are capable to produce at level $s_i = \underline{s}$. U_2 and U_3 have incentives to upgrade their products up to level $s \in (\underline{s}, \bar{s})$ and opportunistically enter the upstream market through negotiation with any downstream producer by making investment. The investment cost functions of upstream and downstream producers $R_U(s)$ and $R_D(s)$ satisfy $R'_U > 0$ and $R'_D < 0$, and $R''_U > 0$ and $R''_D > 0$. These conditions ensure that firms pay less as the agreed sophistication level is closer to their original level and that radical changes are less probable. However, we shall see U_1 sets a price that deters the competitors' entry at equilibrium even if the downstream producer finds the arrangement appealing. We assume that upstream producer

[2]This assumption is justifiable considering the sizable markup on top of the production cost that the leading firm would charge using its monopoly status. Adding costs to the model does not meaningfully affect the following results, which we will briefly address in the discussion section.

can perfectly price discriminate downstream producers. Let p_{ij} and q_{ij} denote the price and the quantity that U_i sells at to D_j. The objective of U_i is to maximize $\sum_j p_{ij}q_{ij}$ subject to downstream producers' optimization problem.

The downstream market is monopolistically competitive where two similar downstream producers D_1 and D_2 vie for the same group of consumers. They optimize their production according to the Cobb-Douglas production function

$$y = AL^{\alpha} \times \left(\overline{k(s)}K\right)^{1-\alpha} \tag{1}$$

where an increasing function $k(s)$ denotes productivity that corresponds to the sophistication level of the input, and $\overline{k(s)}$ the weighted average of productivity by the amount of inputs from each upstream producer in case it has sourced from multiple firms. Formally, $\overline{k(s)} = \left(\sum_i q_{ij}\right)^{-1} \sum_i q_{ij}k(s_i)$. Defining p_{dj} and q_{dj} as the price and the quantity D_j sells at to consumers, the objective of D_j is to maximize $\sum_j p_{dj}q_{dj}$ subject to the consumer optimization problem.

Consumer in the model consists of diverse economic agents ranging from downstream firms of D_j to final consumers in reality, but borrowing from Melitz and Ottaviano (2008) we simplify their aggregate demand into a single utility function

$$U_C = \eta(q_{d1} + q_{d2}) - \frac{1}{2}\left({q_{d1}}^2 + {q_{d2}}^2 + 2\sigma q_{d1}q_{d2}\right) \tag{2}$$

subject to the budget constraint E. $\sigma \in [0, 1]$ and η are parameters of product heterogeneity in the downstream market and the overall consumption level respectively. That is, higher σ indicates products of D_1 and D_2 are highly interchangeable that increased demand for one leads to decreased demand for the other, and higher η implies increased demand for both items.

At the beginning of period 1, U_2 and U_3 make decisions on whether to enter the upstream market through a negotiation with D_1. When the restriction is lifted at $t = 2$, the situation is not the same as with $t = 0$ if either of them has entered a period earlier who U_1 shoud drive out through market competition to restore its monopoly status. Once export restriction is

imposed, firms reorient themselves by maximizing their profit from $t = 1$ to $t = \infty$ minus the investment cost at $t = 1$, if occurred, with all future profits discounted by $\beta \in [0, 1]$ for each period.

2.2 The timing of the game at $t = 0$

All information is complete.

1. U_1 proposes p_{1j}.

2-1. If D_j accepts this proposal, D_j sources only from U_1. D_j decides q_{1j} and sets p_{dj}.

3-1. Consumer decides q_{dj}.

2-2. If D_j rejects this proposal, D_j negotiate with U_2 and U_3 over the sophistication level. They pay $R_D(s)$ and $R_U(s)$ respectively.

3-2. U_i sets p_{ij}.

4-2. D_j decides q_{ij} and sets p_{dj}.

5-2. Consumer decides q_{dj}.

2.3 Equilibrium Analysis

We first consider the case that upstream producers are in full competition. The leading firm extracts sizable rents from its monopoly status in the upstream market as will be shown below, which the potential competitors will be keen to share. Due to this sharp conflict of interests, a strong coalition between the leading firm and the rest is unlikely without any external intervention. Proposition 1 states the pre-restriction Nash equilibrium at $t = 0$ when D_js see lesser players in the upstream market viable.

Proposition 1. *When upstream producers are competitive, the following hold before restriction:*

1. *D_1 can effectively threaten U_1 that it will defect if it sees U_2 and U_3 viable. D_1's gain and U_1's loss from the threat increase as the initial investment cost to harness the competitors R_U and R_D decrease and time discount β increases.*

2. *D_1's gain is always bigger than U_1's loss if such a threat works. Decline of U_1's monopoly status results in greater social efficiency.*

Proof of proposition 1. We first solve the consumer optimization problem. Consumers buy goods from downstream producers until their marginal utility equals price. Therefore, $p_{dj} = \eta - q_{dj} - \sigma q_{d(3-j)}$ for $j \in \{1, 2\}$.

Next, we solve the downstream producers' optimization problem. With wage being externally and competitively determined, marginal return of labor should equal its cost.

$$\frac{\partial y}{\partial L} = \alpha A \left(\frac{\overline{k(s)}K}{L} \right)^{1-\alpha} = w \quad \rightarrow \quad q_{dj} = \left(\frac{w}{\alpha A} \right)^{\frac{\alpha}{1-\alpha}} \times \overline{k(s)}K \equiv \frac{1}{\gamma} \times \overline{k(s)}K$$

Total cost is given as $wL_0 + \overline{p_{\cdot j}}K_0$ where L_0 and K_0 denote the amount of inputs at equilibrium. Note that $wL_0 = \alpha L^{\alpha} \left(\overline{k(s)}K \right)^{1-\alpha} = \alpha q_{dj}$ and $\overline{p_{\cdot j}}K_0 = \sum_i p_{ij} q_{ij}$. D_j determines q_{dj} where its profit is maximized, so for all i such that $q_{ij} > 0$,

$$\frac{\partial \Pi_j}{\partial q_{ij}} = \frac{\partial}{\partial q_{ij}} \left[q_{dj} \left(p_{dj} - \alpha \right) - \sum_k p_{kj} q_{kj} \right] = \frac{k(s_i)}{\gamma} \left(p_{dj} - \alpha - q_{dj} \right) - p_{ij} = 0.$$

We learn from above that the effective price $\frac{p_{ij}}{k(s_i)}$ is the same across all inputs if D_j sources from multiple upstream firms. Imposing symmetry between D_1 and D_2, or equivalently $p_{d1} = p_{d2}$, $q_{d1} = q_{d2}$, $c_1 = c_2$, we obtain

$$q_{dj} = \frac{\eta - \alpha - \gamma c_j}{2 + \sigma}, \;\; p_{dj} = \eta - (1 + \sigma) q_{dj}, \;\; \Pi_j = (q_{dj})^2. \tag{3}$$

If we drop the symmetry,

$$q_{dj} = \frac{(2 - \sigma)(\eta - \alpha) - 2\gamma c_j + \sigma \gamma c_{3-j}}{4 - \sigma^2}, \; p_{dj} = \eta - q_{dj} - \sigma q_{d(3-j)}, \; \Pi_j = (q_{dj})^2. \tag{4}$$

If U_1 is the only available supplier for downstream suppliers, U_1 maximizes its profit by choosing p_1 that maximizes $p_1 q_1$ according to (3). Therefore,

$$p_{11}^m = \frac{k(\bar{s})(\eta - \alpha)}{2\gamma}, \; q_{d1}^m = \frac{\eta - \alpha}{2(2 + \sigma)}.$$

Meanwhile, if D_1 considers U_2 and U_3 viable suppliers, D_1 can threaten U_1 to lower price or they will defect and get supplies from the competitors at a lower price. U_1 should tolerate a lower price to maintain its monopoly, so their entry might as well be beneficial to U_1. But it is always not the case.

Suppose that U_2 and U_3 reached an agreement with D_1 at the sophistication level $s^o \in (\underline{s}, \bar{s})$ without necessarily excluding provision from U_1. Let p^o_{i1} and q^o_{i1} be the price and the quantity at equilibrium. Their entry is easily deterred by the proposal U_1 makes to D_1 *before entry* to provide $q^d_{11} + q^o_{21} + q^o_{31}$ units of the goods at $p^o_{11}q^o_{11} + \frac{k(s^o)}{k(\bar{s})}\big(p^o_{21}q^o_{21} + p^o_{31}q^o_{31}\big)$. This proposal is pareto efficient for U_1 and D_1 since U_1 earns $\frac{k(s^o)}{k(\bar{s})}\big(p^o_{21}q^o_{21} + p^o_{31}q^o_{31}\big)$ more and D_1 is supplied with higher-quality goods of the same amount at a lower cost. Also, this proposal satisfies D_1's optimization stated in (4) since the proposal preservs the effective price c_1 that D_1 pays for each unit. The implication of the proposal is that whatever U_2 and U_3 offers to D_1, there is at least one proposal that is more efficient to both U_1 and D_1 among many pre-entry proposals that U_1 can make to D_1, and U_1's dominance is always guaranteed.

U_1 then maximizes its profit while guaranteeing D_1 at least the profit it could have earned by contracting with U_2 and U_3. Imagine U_2 and U_3 try to enter the market. The competition between them will lead their profit to zero, and the price will be determined at the level that barely recovers their initial investment cost $R_U(s)$ through infinite periods. However, knowing that U_1 can undercut their price and they will never be able to recover the investment cost unless they set the price at zero, the only way D_1 can keep the defection threat credible is to fully assume U_2 and U_3's investment cost if such arrangement is possible. D_1 does not need to invest for both firms, so we posit $p_{21} = 0$, $q_{21} > 0$, $q_{31} = 0$. q^o_{d1} is determined as

$$q^o_{d1} = \frac{(2-\sigma)(\eta-\alpha) + \sigma\gamma\frac{p^m_{11}}{k(\bar{s})}}{4-\sigma^2} = \frac{(4-\sigma)(\eta-\alpha)}{2(4-\sigma^2)}.$$

D_1's final payoff becomes $\frac{\Pi_1}{1-\beta} - R_D(s) = \frac{(q^o_{d1})^2}{1-\beta} - R_U(s) - R_D(s)$, based on which D_1 solely decides s^o since U_2's payoff is always zero. Returning to the negotiation between U_1 and D_1, U_1 makes a proposal that maximizes $p_{11}q_{11} = \frac{p_{11}q_{d1}}{k(\bar{s})}$ under the constraint

$$(q_{d1})^2 \geq \max\big[\max_s\big[(q^o_{d1})^2 - (1-\beta)\big(R_U(s) + R_D(s)\big)\big],\ (q^m_{d1})^2\big]. \tag{5}$$

We know that $p_{11}q_{d1}$ decreases in q_{d1} from (4), so q_{d1}^d attains the equality in (5).

1. Payoff is proportional to the profit in each period, so v_{d1}^d is greater than v_{d1}^m by (5) and v_{u1}^d is equal to v_{u1}^m only when $(q_{d1}^d)^2 = (q_{d1}^m)^2$. Decrease in R_U and R_D and increase in β lead to a higher q_{d1}^d.
2. It is enough to show $\Delta(q_{d1})^2 - \Delta(p_{11}q_{11}) > 0$. Using $\Delta x^2 = 2x\Delta x + (\Delta x)^2$, a simple manipulation leads to

$$\left(3 - \frac{\sigma^2}{2}\right)\Delta q_{d1}\left(\Delta q_{d1} + \frac{(\eta - \alpha)(4 - \sigma^2 - 2\sigma)}{2(2+\sigma)(6-\sigma^2)}\right) > 0,$$

which is always true as long as $\Delta q_{d1} > 0$. ■

Proposition 1 shows that D_1 can undermine U_1's monopoly status by leveraging the existence of its competitors. The results are intuitive. D_1 has a choice to defect and have either U_2 or U_3 (not both) supply for it at a lower cost, and if that proves unprofitable D_1 just chooses to stay. U_1 would have tolerated their market entry rather than lowering the price otherwise, but it finds better to guarantee D_1 at least the profit D_1 could have earned by contracting with other suppliers and to maintain its monopoly. This is because U_1 is capable to offer at whatever price and quantity that U_2 and U_3 offe *before* their entry and preempt their profits. However, for such a threat to be credible, D_1 should assume all investment costs of the competitors, or U_1 will undercut their pricing to secure even a bit of profit and the competitors will never be able to recover the investment costs. If upgrade costs are trivial, U_1 faces perfect competition with other suppliers who are ready to supply almost at no cost competing each other. In this case, U_1 may lose all profits coming from D_1 and D_1's profit grows four to nine times depending on the level of σ. Redistribution in this model only occurs between U_1 and D_1, and decline in U_1's profit never exceeds growth in D_1's since monopoly is socially inefficient.

Figure 2 visualizes the above proposition. Suppose that the equilibrium price is formed at p_{11}^m under U_1's pure monopoly. When D_1 defects to U_2, D_1 can enjoy the surplus represented by the shaded region on the right panel as the price is formed at a level that barely covers U_2's initial investment costs $R_U(s^o)$, the area of the rectangle below the equilibrium, and R_1

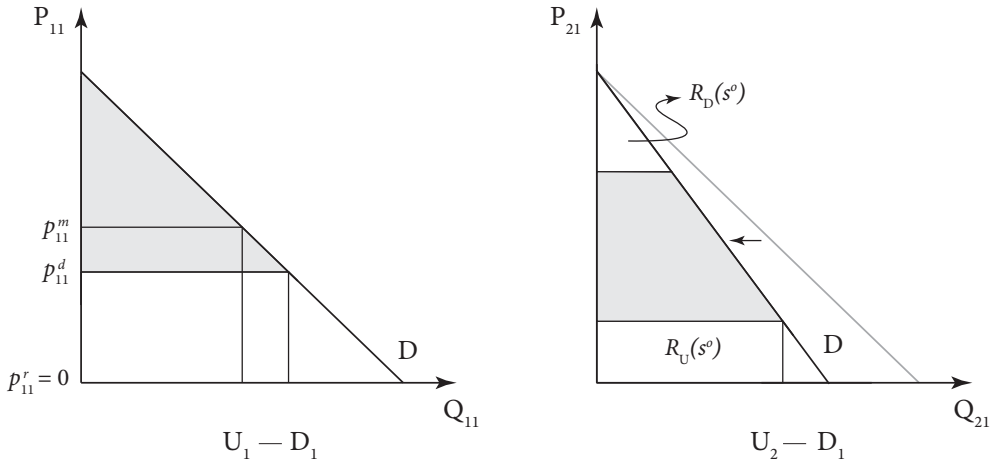

Figure 2: Price determination in the competitive upstream market: each panel represents the demand function of D_1 for the current supplier U_1 and the potential supplier U_2. The shaded region on the right denotes the surplus D_1 enjoys when it defects to D_2. U_1 has to then drive down its price from p_{11}^m to p_{11}^d where D_1 enjoys as much surplus.

pays its own investment costs $R_D(s^o)$, the area of the upper triangle, where s^o denotes their agreed sophistication level between two. For U_1 not to lose entire profits, it ensures D_1 as much surplus by driving down the price from p_{11}^m to p_{11}^d until the areas of the two shaded regions are equal. This is always possible since D_1 always prefers inputs from U_1 over those from U_2 when the prices are the same, so D_1's demand curve for U_2 is inside that for U_1. The equilibrium occurs at the point where the area of the shaded region on the left panel is equal to the surplus D_1 can enjoy when D_1 assumes all of the U_2's investment costs and defects.

This result implies that the mere presence of lurking export restriction might be detrimental to the exporter. When and how D_1 sees the competitors viable is not explicitly accounted in this model, but

Here, the anticipation of export restriction may trigger the downstream supplier's greater business strategy. D_1 would always have liked to bypass U_1's monopoly but it entails risks and uncertainty to change its already well-proven manufacturing procedures. Escalating political tension within and without the supplier country, for example, might nudge D_1 to look for alternatives and take those risks while allowing D_1 to use it as a card in negotation with the monopolizing supplier. However, this never amounts to actual source diversification

of D_1 before the tension develops into an actual diplomatic crisis. When the market foresees export restriction is around the corner, difference should arise in response capability across downstream producers. More capable producers are expected to better afford investment cost R_D better thanks to the technological superiority and better access to capital, and therefore a bigger loss for U_1.

Next, we consider when export restriction takes place at $t = 1$. The restriction renders U_1 unable to make a counterproposal to deter the competitors' entry. Proposition 2 describes the permanent equilibrium shift that occurs from this one-time shock.

Proposition 2. *When upstream producers are competitive, the following hold after restriction:*

1. *U_1 suffers a moderate loss if D_1 fails at securing alternative sources and loses more than profits coming from D_1 otherwise, while D_1 suffers a sizable loss if D_1 fails at doing so and makes a modest gain otherwise. D_1 always secures alternative sources if it successfully threatened U_1 before the restriction.*

2. *As the export restriction protracts, D_1 more likely succeeds in negotiating with the alternative suppliers and their investment costs are less likely transferred to D_1. U_1 is more likely to lose while D_1 may either gain or lose.*

Proof of proposition 2. At $t = 2$, all suppliers set the price at zero, and $q_{d1} = q^o_{d1}$. The equilibrium at $t = 1$ then occurs where D_1's margin is maximized under the constriant that U_2 and U_3's payoff are non-negative. Without loss of generality we set $q_{31} = 0$, $p_{21} = \frac{R_U(s)}{q_{21}} = \frac{k(s)R_U(s)}{q_{d1}}$. Let q^r_{d1} be the quantity under restriction. q^r_{d1} is the bigger solution of the following equation:

$$q^r_{d1} + \frac{2\gamma R_U(s)}{4-\sigma^2}\frac{1}{q^r_{d1}} = \frac{(4-\sigma)(\eta-\alpha)}{2(4-\sigma^2)}. \tag{6}$$

D_1's payoff is given as $(q^r_{d1})^2 + \frac{\beta(q^o_{d1})^2}{1-\beta} - R_D(s)$. Compared to (5), $R_U(s)$ here is implicit in q^r_{d1} which allows D_1 to have the downstream producer pay for part of its own investment cost.

1. If D_1 fails at securing alternative sources, D_1 does not produce any goods and resumes after the export restriction is lifted. If D_1 manages to do so, U_1 cannot make profits even after the export restriction is lifted. For D_1, notice that D_1's payoff is the same as the pre-restriction payoff under the defection threat if D_1 and U_2 agree to transfer the investment cost to D_1 again. However, in this case D_1 does not need to assume all the cost. Note that the payoff is concave in $R_U(s)$ and

$$\frac{\partial}{\partial R_U(s)}\left[(q_{d1}^r)^2 + \frac{\beta(q_{d1}^o)^2}{1-\beta} - R_D(s)\right]_{R_U(s)=0} = 1 - \frac{4\gamma}{4-\sigma^2}$$

when D_1 can freely allocate $R_U(s)$ and $R_D(s)$ under the constraint that their sum is constant. Therefore, the cost is shifted towards U_2 if the derivative is positive.

2. η and β are affected as the export restriction protracts. D_1's decision on sourcing diversification depends on its payoff differential, or $(q_{d1}^r)^2 - R_D(s) + \frac{\beta}{1-\beta}\big((q_{d1}^o)^2 - (q_{d1}^d)^2\big)$. This payoff differential is increasingly decided by the first two terms as β decreases. Increase in η leads to increase in q_{d1}^r, so D_1 is more likely to source from U_2. The investment cost allocation problem is defined as

$$\frac{\partial}{\partial R_U(s)}\big((q_{d1}^r)^2 + R_U(s) - (R_U(s) + R_D(s))\big) = 0,$$

or equivalently, $\psi q(\eta) = q_{d1}^r$ where $q(\eta) \equiv \frac{(4-\sigma)(\eta-\alpha)}{2(4-\sigma^2)}$ and ψ is a constant independent of η. Plugging this in (6), we obtain $R_U(s) \propto (q(\eta))^2$. Therefore, as η increases the investment cost is shifted towards U_2 at every level of s.

For D_1's payoff, refer to the simulation study. ■

Proposition 2 states that U_1 always loses from export restriction and D_1 may gain or lose depending on the value of the investment costs R_U and R_D. U_1 has no way to increase its revenue cutting off economic ties with D_1. If U_2 enters the market, U_1 loses the remaining profits from D_1 plus some of the profits from D_2 who now exploits U_2's increased sophistication level to lower the input price. On the other hand, D_1 is in an even more favorable position since it can not only implement what it threatened U_1 to do at $t = 0$ but shift some of the investment costs to U_2 who now can afford a non-zero price during the restriction,

which allows D_1 to secure alternative sources even when it is unwilling or unable to bear the competitors' investment costs.

Another interesting point is when the restriction lengthens. The common wisdom is that downstream producers would feel more pressurized the longer the export restriction becomes. Proposition 2 says it is not always the case. Downstream producers respond to the restrictive measure by investing in alternative sources, and those costs can be recovered through a longer period as the restriction is extended. This did not matter at $t = 0$ since U_2 was unable to make any profits whatsoever because of U_1's counterproposal, which had D_1 assume all of U_2's cost. Meanwhile, under the restriction, U_2 can raise revenue just as much to offset the investment cost and D_1 takes advantage of the lower input price thanks to the extension with the level of the agreed sophistication level held constant. D_1 weighs additional profits it will earn during and after the restriction against the upgrade cost when making the investment decision at the beginning of the restriction, so β governs the entire process. We assume that β is small enough for D_1 not to exaggerate the post-restriction profit increments.[3] As the restriction protracts, D_1 faces a better chance to run profits well above the initial investment costs and is more likely better off by curtailing the massive transfer to U_2 and instead tolerating the price raise.

Can export restriction discipline downstream producers and their countries? Not always, says this model, and there is even a possibility of backfiring when D_1 is a global giant. It makes sense for U_1's country to impose a restriction only when it brings immediate external benefits that outweigh the loss U_1 suffers plus when D_1 is unable to even partly overcome the technological gap and harness lesser players in the upstream market. However, the firm being targeted has to be economically and politically sizable for the restriction to exert utmost pressure against the targeted country. This model shows how export restrictions are unlikely to settle in as a new mode of economic warfare between advanced countries. Governments would willingly take their time until they see their targets crumble under pressure – only if

[3]There are two lines of reasoning for this. First, global corporates are expected to show good numbers to investors every quarter while both governments, being conscious of their own citizens, will not easily back down. Lasting export restriction translates to low β in this model as it elongates the span of each period. Second, in reality the alterantive suppliers will likely be driven out from the upstream market in the end due to the cost and technological advantage the market leader enjoys unless they have enough time to build up their own strengths over the restriction. Therefore, D_1 is probably not going to enjoy as high profits after a few periods even if its β is close to 1.

time is on their side. The longer they stand in the way, the more successful those affected firms will be in finding a way around by demolishing the monopoly status of their supplier and enjoying much higher profits for good.

So far, we have characterized the equilibria under the assumption that upstream producers are competitive. We now consider the case when they are collusive. Although the leading firm and the potential competitors had no common interest prior to restriction, the leading firm might be interested in the arrangement to share its profits with them in return for a share of theirs from the restriction on. The competitors might also favor such a proposal since they are unable to run any profits while competing with each other. This can happen as the leading firm takes over other smaller foreign companies as a way to bypass the restriction at the firm level, or the exporting government proposes side payment to other countries to have them join the restrictive measure at the political level. Suppose that U_1 takes $\theta \in (0, 1)$ of the total payoff U_is raise since $t = 1$, and U_2 and U_3 keep each half of the rest. Proposition 3 illustrates the post-restriction collusive equilibrium.

Proposition 3. *When upstream producers are collusive, D_1 always loses* all *profits during the restriction while U_1 experiences a moderate loss as long as R_D is not prohibitively high. D_1 is indifferent between sourcing from U_1 and from its competitors, and earns the same profit as before when the restriction is lifted.*

Proof of proposition 3. If the competitors do not enter the market at $t = 1$, the equilibrium at $t = 2$ is the same as at $t = 0$. Suppose otherwise and the competitors produce at the level s. For upstream producers to maximize their aggregate payoff, only one of U_2 and U_3 enters the market. Assuming U_2 entered at $t = 1$, the producers maximize $p_{11}q_{11} + p_{21}q_{21} = c_1 q_{d1}$ at $t = 2$. Their optimization problem is reduced to deciding the level of c_1, which is equivalent to the problem at $t = 0$. If the level of c_1 is the same with that at $t = 0$, they are indifferent to who supplies to D_1 how much and so is D_1. D_1's profit is the same as before.

Returning to at $t = 1$, upstream producers' decision at this point does not affect their later payoffs. They will therefore make D_1 pay as much investment cost as possible, and the agreed level of sophistication level occurs where $(q_{d1}^m)^2 - R_D(s) = 0$. D_1 loses all profit for one period and the upstream producers collectively bear an additional cost of R_U. U_1's

payoff is given $\theta\big(\frac{p_{11}^m q_{11}^m}{1-\beta} - R_U(s)\big)$. ■

According to Proposition 3, U_1 still exerts a power tantamount to monopoly via lesser players even when barred from direct export to D_1, whereas the targeted firm loses whatever profits it had enjoyed before during the export restriction. Since the upstream producers will ask D_1 to pay for the investment cost as much as it can afford at $t = 1$, the whole consumer surplus of D_1 is consigned to them. Time is also on their side in this case for it will help D_1 make bigger investment for the benefit of U_js. The market returns to the pre-restriction equilibrium after the restriction is lifted besides that the once competitors sell their products at the price corresponding to U_1's monopolistic price vis-à-vis the technological gap and their entry does not contribute any to D_1's welfare. Upstream producers' aggregate revenue is constant throughout these periods. The political implication is the opposite here: the more capable downstream producers are, the more likely export restrictions are. The targeted firms would try to address the external causes that triggered the restriction rather than seek solutions within the market, and their government would more seriously consider bending to the exporting government.

3 Discussion

The main takeaway is that the coalition between producers matters in trade restriction. Whether the imposer can achieve its goals through restriction depends on if it manages to elicit full cooperation with other producers in the market. This result is in line with that of the economic sanctions literature that, for a sanction to be successful, the proposer should ensure other countries to join it so that the sanctioned country cannot easily walk around it (Martin, 1994).

They do not enter if the exporter strikes a collusive negotiation with them or the government persuades other countries to join the restrictions. The two have divergent preferences over the options of collusion and multilateral restrictions, but the government has few economic interests in deterring the exporter from continuing operation when it fails to mobilize other countries' participation as in the Korea-Japan case.

adding costs

illustration: The second study qualitatively examines how Samsung Electronics and its suppliers fared during the Korea-Japan trade row. In August 2019, Japan banned exports of several industrial materials to Korea that were crucial in chipmaking and Japan was virtually monopolizing. Being aware that the measure would hurt its own firms, Japan was believed to try to overturn the recent decision of the Korean Supreme Court by bringing the giant chipmaker to its knees that alone makes up a significant portion of national exports. However, Samsung surprised Tokyo announcing it would tap into alternative sources in Russia immediately and succeeded in local production three months later. Some of the Japanese firms continued provision via foreign partners and others belatedly built plants within Korea, both bypassing the regulations that have yet to be lifted officially until now. This demonstrates the main results of our theory: the implicit reliance is not the same with the explicit reliance, the targeted firms trade costs with quality of inputs, and export regulations are meaningful only when the government is willing to further enforce domestic firms.

Future directions.... Dynamic model: s varies over time -¿ cost/technological advantage Network? multiplayers. labor?

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A Variables that determine the model structure

We identify three groups of variables that affect the structure of the model. The first group relates to the banned goods themselves. The players' objectives depend on the role technology plays in their production. If it has an intrinsic value, it facilitates the production process of downstream producers but does not directly add values to consumers. Purer industrial materials such as rare earth or hydrofluoric acid allow cost reduction by boosting production yields, but consumers would feel indifferent if a producer chose a tighter quality control over purer materials. Conversely, if it has an extrinsic value, it directly speaks to consumers just as higher-performing chips in a machine better appeal to consumers and command higher value addition. Another issue is how much differentiated the banned goods are from the competing goods in the market. Large heterogeneity implies significant entry barriers for latent players and hampers technology transfer from one area to another. The case reduces to the conventional hold-up problem under extreme heterogeneity.

The second group addresses the market environment. The market competitiveness is decided by not only the number of active and latent competitors but the relationship among them. We will see below that the effects of trade restriction starkly differ when they are competitive and when they are collusive. With the market competition held constant, the more upstream the banned goods sit in the supply chain, the easier it would be for downstream producers to secure alternatives and the harder for upstream producers. This is partly because product heterogeneity increases down the production stages, but also because downstream producers would have less to consider when looking for alternatives to replace the upper part of the chain as the broken leg moves upwards. Whether the restriction preserves or upends the status quo also matters. Figure 1 describes a restriction that intends to harm the incumbent player, but restrictions could instead deter entry of a new player like the U.S. trying to curb China's rise in the advanced semiconductor industry.

The last group includes variables that provide the political context of the restriction. Whether it is the upstream country or the downstream country who initiated the restriction would matter in studying driving forces behind the measure, but not so much in studying its distributional effect. Domestic politics rather matters in that questions like who initiated it and how much determined they are give hints on its duration. In democratic states, for example, legislation is more inflexible than executive orders and therefore legislative-imposed restrictions would last much longer. Leaders' determination, along with the nature of the trade dispute, affects the market environment by drawing boundaries of what firms can do and cannot. If a state manages to mobilize other states to join the restriction and has a strong command over its firms, the targeted firms would be less successful in eluding negative consequences. This includes all kinds of diplomatic means like secondary boycott, normative persuasion and side payments.